Curated Intermediate Worksheet

Chapter 1: Integers (Topics up to Exercise 1C)

Time Allowed: 60 Minutes — Maximum Marks: 40

General Instructions:

1. This worksheet is divided into two sections: **Section A** (Multiple Choice Questions) and **Section B** (Subjective Problems).
2. **Section A** contains 10 MCQs of 1 mark each. Questions 9 and 10 are High Order Thinking Skills (HOTS) questions.
3. **Section B** contains 10 subjective problems of 3 marks each. Problems 9 and 10 are High Order Thinking Skills (HOTS) problems.
4. Detailed, step-by-step solutions are provided at the end of the worksheet for self-assessment.

SECTION A: Multiple Choice Questions (10 Marks)

Q1. If a and b are integers such that $a \cdot b < 0$ and $a + b < 0$, which of the following must be true?

(a) $a > 0$ and $b > 0$
(b) $a < 0$ and $b < 0$
(c) The integer with the larger absolute value is negative
(d) The integer with the larger absolute value is positive

Q2. How many integer values of x satisfy the equation $\big||x| - 5\big| = 2$?

(a) 2
(b) 4
(c) 3
(d) 0

Q3. If p is a negative integer, then the predecessor of the successor of p is:

(a) $p + 1$
(b) $p - 1$
(c) p
(d) $-p$

Q4. If a and b are non-zero integers such that $a \div b = b \div a$, which of the following is correct?

(a) $a = b$
(b) $a = -b$
(c) $a = \pm b$

(d) $a \cdot b = 1$

Q5. The value of $(-45) \times 102$ is equal to:

(a) $(-45) \times 100 + 2$
(b) $(-45) \times 100 - 90$
(c) $(-45) \times 100 + (-45) \times 2$
(d) Both (b) and (c)

Q6. If $x < y < 0$, which of the following expressions yields the largest value?

(a) $x - y$
(b) $y - x$
(c) $x + y$
(d) $2x$

Q7. The sum of three consecutive odd integers is -57. The smallest of these three integers is:

(a) -21
(b) -17
(c) -19
(d) -23

Q8. How many integers satisfy the inequalities $|x| \leq 3$ and $x > -2$ simultaneously?

(a) 6
(b) 5
(c) 4
(d) 3

HOTS (High Order Thinking Skills)

Let a, b, and c be non-zero integers such that $a \cdot b \cdot c < 0$, $a \cdot b > 0$, and $a < c$. Which of the following is true about their signs?

Q9. (a) $a < 0$, $b < 0$, $c > 0$
(b) $a < 0$, $b < 0$, $c < 0$
(c) $a > 0$, $b > 0$, $c < 0$
(d) Cannot be determined uniquely

HOTS (High Order Thinking Skills)

Q10. Let a and b be integers such that $|a| \times |b| = 12$. If the distance between a and b on the number line is as small as possible, what is the value of $a + b$ if $a < 0 < b$ and $|a| > b$?

(a) 7

(b) 1

(c) -1

(d) -7

SECTION B: Subjective Problems (30 Marks)

Each question carries 3 marks. Answer with complete mathematical steps and justifications.

P1. Simplify the following multi-step integer expression by showing each intermediate step:
$[(-15) \times (-4) \times (-3)$
$\div [(-12) \times (-5)]$
$]$

P2. Find all negative integers x that satisfy the modulus inequality $|x - 3| < 4$. If no such integers exist, provide a detailed algebraic proof.

P3. Verify the Distributive Property of Multiplication over Addition for the integers $a = -12, b = -8$, and $c = 15$. Show the step-by-step evaluation for both the Left-Hand Side (L.H.S.) and the Right-Hand Side (R.H.S.).

P4. Find the product of the predecessor of the smallest positive integer and the successor of the largest negative integer. Clearly define each integer term in your solution.

P5. A chemical solution is kept in a laboratory at 15°C. A scientist cools the solution at a constant rate of 4°C per hour.

(a) What will be the temperature of the solution after 8 hours?

(b) If the freezing point of the solution is -20°C, will the solution freeze within these 8 hours? Justify your answer.

P6. A submarine is at a depth of 450 m below sea level. It begins to ascend toward the surface at a constant rate of 15 m per minute.

(a) Express its initial position and ascending rate as signed integers.

(b) Find its position relative to sea level after 25 minutes of ascending.

P7. A cement company earns a profit of Rs. 8 per bag of white cement sold and a loss of Rs. 5 per bag of grey cement sold.

(a) The company sells 3,000 bags of white cement and 5,000 bags of grey cement in a month. What is its overall profit or loss?

(b) What is the minimum number of white cement bags the company must sell to have neither profit nor loss, if it sells $6,400$ bags of grey cement?

P8. A competitive exam consists of 30 questions. For each correct answer, $+4$ marks are awarded; for each incorrect answer, -2 marks are awarded; and for each unattempted question, -1 mark is awarded. Ananya answered 15 questions correctly and scored a total of 40 marks. Assuming she attempted some questions incorrectly and left the rest unattempted, find the number of questions she attempted incorrectly and the number she left unattempted.

HOTS (High Order Thinking Skills)

P9. Let p and q be integers satisfying the inequalities $|p| \leq 5$ and $|q| \leq 4$.

(a) Find the maximum possible value of the expression $(p - q)$.

(b) Find the minimum possible value of the product $(p \times q)$.

(c) Calculate the difference when the minimum product from (b) is subtracted from the maximum difference from (a).

HOTS (High Order Thinking Skills)

P10. Let a, b, and c be three distinct non-zero integers such that a divides b $(a \mid b)$, and b divides c $(b \mid c)$.

(a) Prove algebraically that a must divide c $(a \mid c)$.

(b) If $a \times b \times c = -64$, find a set of distinct non-zero values for (a, b, c) that satisfy all the given conditions.

SECTION C: Detailed Solutions Key

Part 1: Multiple Choice Questions (Solutions)

Q1. Correct Option: (c)
Since $a \cdot b < 0$, the two integers must have opposite signs (one positive, one negative). Since their sum $a + b < 0$, the negative integer must have a larger absolute value than the positive integer (e.g., $-5 + 3 = -2 < 0$, where $|-5| > |3|$). Thus, the integer with the larger absolute value is negative.

Q2. Correct Option: (b)
Let $\big||x| - 5\big| = 2$. This gives two cases:

- **Case 1:** $|x| - 5 = 2 \implies |x| = 7 \implies x = \pm 7$.
- **Case 2:** $|x| - 5 = -2 \implies |x| = 3 \implies x = \pm 3$.

The set of integer solutions is $\{-7, -3, 3, 7\}$. There are exactly 4 solutions.

Q3. Correct Option: (c)
The successor of an integer p is $(p + 1)$. The predecessor of any integer y is $(y - 1)$.
Therefore, the predecessor of the successor of p is:
(p + 1) - 1 = p

This is true for all integers, negative or positive.

Q4. Correct Option: (c)
Given $a \div b = b \div a \implies \frac{a}{b} = \frac{b}{a}$. Since a and b are non-zero integers:
a$^2 = b^2 \implies a = \pm b$

This means the absolute values of a and b are equal, so either $a = b$ or $a = -b$.

Q5. Correct Option: (d)
By the Distributive Property of multiplication over addition, $x \times (y + z) = x \times y + x \times z$.
We can express $(-45) \times 102$ as:
(-45) $\times(100 + 2) = (-45) \times 100 + (-45) \times 2$ (Option c)

Evaluating this: $-4500 + (-90) = -4590$.
Now let's check Option (b): $(-45) \times 100 - 90 = -4500 - 90 = -4590$.
Since both expressions evaluate to the same correct product of -4590, both (b) and (c) are correct.

Q6. Correct Option: (b)
Since $x < y < 0$, both are negative, and $|x| > |y|$.

- $x - y$: negative (since $x < y$).
- $y - x$: positive, because $y > x \implies y - x > 0$. Specifically, $y - x = |x| - |y| > 0$.
- $x + y$: negative (sum of two negative integers).
- $2x$: negative.

The only positive value is $y - x$, which is therefore the largest.

Q7. Correct Option: (a)

Let the three consecutive odd integers be n, $n+2$, and $n+4$ (where n is an odd integer). Their sum is:

n + (n + 2) + (n + 4) = -57 $\implies 3n+6=-57 \implies 3n=-63 \implies n=-21$

The three integers are $-21, -19, -17$. The smallest of these is -21.

Q8. Correct Option: (b)

The inequality $|x| \leq 3$ represents all integers from -3 to 3: $\{-3, -2, -1, 0, 1, 2, 3\}$.
Applying the second condition $x > -2$, we filter out integers less than or equal to -2.
The remaining integers are $\{-1, 0, 1, 2, 3\}$, which counts to exactly 5 integers.

Q9. Correct Option: (b)

We analyze the signs step-by-step:

- Since $a \cdot b > 0$, a and b must have the same sign (both positive or both negative).
- Since $a \cdot b \cdot c < 0$ and we know $(a \cdot b)$ is positive, we must have $c < 0$ (positive $\times$ negative = negative).
- Since $c < 0$ and we are given $a < c$, then a must also be negative (less than a negative number is negative).
- Since $a < 0$ and a, b have the same sign, b must also be negative.

Thus, all three integers are negative: $a < 0, b < 0, c < 0$.

Q10. Correct Option: (c)

Since $a < 0 < b$, a is negative and b is positive, so $|a| = -a$ and $|b| = b$.
We are given $|a| \times |b| = 12 \implies (-a) \times b = 12$.
The distance between a and b on the number line is $b - a = |b| + |a|$.
To minimize this distance, we look for factor pairs of 12 whose sum $|a| + |b|$ is minimized:

- Factor pairs of 12 are $(1, 12)$, $(2, 6)$, and $(3, 4)$.
- The sums of these pairs are $1 + 12 = 13$, $2 + 6 = 8$, and $3 + 4 = 7$.
- The minimum sum is 7, which corresponds to $\{|a|, b\} = \{3, 4\}$.

We are given the condition $|a| > b$, which means $|a| = 4$ and $b = 3$.
Since $a < 0$, we have $a = -4$.
Therefore, $a + b = -4 + 3 = -1$.

Part 2: Subjective Problems (Solutions)

P1. Solution:

We evaluate the expression step-by-step:

- **Step 1: Evaluate the first bracket (Numerator):**

(-15) $\times(-4) \times (-3)$

First, multiply the first two terms: $(-15) \times (-4) = 60$ (since negative $\times$ negative = positive).
Next, multiply by the third term: $60 \times (-3) = -180$ (since positive $\times$ negative = negative).

- **Step 2: Evaluate the second bracket (Denominator):**

(-12) $\times(-5) = 60$ (since negative $\times$ negative = positive)

- **Step 3: Perform the division:**

60 = -3 (since negative ÷ positive = negative)

Final Answer: -3

P2. Solution:

To solve the inequality $|x - 3| < 4$, we write it as a compound inequality:
¡ x - 3 ¡ 4

Adding 3 to all parts of the inequality to isolate x:
+ 3 ¡ x ¡ 4 + 3 $\implies -1 < x < 7$

The integers that satisfy this inequality are $\{0, 1, 2, 3, 4, 5, 6\}$.
None of these integers are negative (since 0 is neither positive nor negative, and $1, 2, 3, 4, 5, 6$ are positive).
Therefore, there are **no negative integers** that satisfy the inequality.
The solution set for negative integers is empty ($\emptyset$).

P3. Solution:

We need to verify $a \times (b + c) = (a \times b) + (a \times c)$ for $a = -12, b = -8$, and $c = 15$.

- **Left-Hand Side (L.H.S.):**

L.H.S. = a $\times(b + c) = -12 \times (-8 + 15)$

First, evaluate the addition inside the parentheses: $-8 + 15 = 7$.
Now multiply: $-12 \times 7 = -84$.

- **Right-Hand Side (R.H.S.):**

R.H.S. = (a $\times b) + (a \times c) = [(-12) \times (-8)$
+ $[(-12) \times 15]$
$]Evaluatethefirstproduct$:(-12) $\times(-8) = 96$.
Evaluate the second product: $(-12) \times 15 = -180$.
Now perform the addition: $96 + (-180) = 96 - 180 = -84$.

Since L.H.S. = R.H.S. = -84, the Distributive Property is verified.

P4. Solution:

Let's identify the integers based on their definitions:

- The smallest positive integer is 1.
- The predecessor of an integer x is $x - 1$. So, the predecessor of 1 is:

- 1 = 0

- The largest negative integer is -1.
- The successor of an integer y is $y + 1$. So, the successor of -1 is:

+ 1 = 0

Now, find their product:

0 = 0

Final Answer: 0

P5. Solution:

Let the initial temperature be +15°C and cooling rate be −4°C per hour.

(a) **Temperature after 8 hours:** The change in temperature after 8 hours is:

(-4°C) = -32°C

The final temperature is:

°C + (-32°C) = -17°C

(b) **Will it freeze?** The freezing point is −20°C.

Comparing the final temperature and the freezing point:

°C > -20°C

Since the temperature of the solution (−17°C) is warmer than its freezing point (−20°C), the solution **will not freeze** within these 8 hours.

P6. Solution:

(a) **Signed Integers Representation:**

- Depth below sea level is represented as negative: Initial Position = −450 m.
- Ascending (moving upward) is represented as positive: Ascending Rate = +15 m/min.

(b) **Position after 25 minutes:** The total change in depth after 25 minutes is:

min × 15 m/min = +375 m

The final position relative to sea level is:

m + 375 m = -75 m

Thus, the submarine is at a position of **−75 m relative to sea level** (or at a depth of 75 m below sea level).

P7. Solution: Let profit per white cement bag be +Rs. 8, and loss per grey cement bag be −Rs. 5.

(a) **Overall Profit or Loss:** Profit from selling 3,000 bags of white cement:

8 = Rs. 24,000

Loss from selling 5,000 bags of grey cement:

(-5) = -Rs. 25,000

Net profit/loss is:

+ (-25,000) = -Rs. 1,000

Since the result is negative, the company incurs an overall **Loss of Rs. 1,000**.

(b) **White bags needed for zero profit/loss:** Let w be the number of white cement bags to sell. For a net balance of 0:

w × 8 + 6,400 × (−5) = 0

w - 32,000 = 0 $\implies 8w = 32,000 \implies w = 4,000$

The company must sell **4,000 bags of white cement** to have neither profit nor loss.

P8. Solution:

Let the total number of questions be 30. Ananya answered 15 correctly.

Let x be the number of incorrect questions, and y be the number of unattempted questions.

The sum of questions gives:

+ x + y = 30 $\implies x + y = 15 \implies y = 15 - x$ — (Equation 1)

Using the scoring system, her total score is 40 marks:

(+4) + x ×$(-2) + y \times (-1) = 40$

Simplify this equation:

- 2x - y = 40 $\implies 2x + y = 20$ — (Equation 2)

Substitute $y = 15 - x$ from Equation 1 into Equation 2:

x + (15 - x) = 20 $\implies x + 15 = 20 \implies x = 5$

Now solve for y:

y = 15 - 5 = 10

Therefore, Ananya attempted **5 questions incorrectly** and left **10 questions unattempted**.

P9. Solution: We are given the bounds $-5 \le p \le 5$ and $-4 \le q \le 4$.

(a) **Maximum possible value of $(p - q)$:** To maximize the difference $(p - q)$, we choose the maximum possible value of p and the minimum possible value of q:

Max(p) = 5, Min(q) = -4

Max(p - q) = 5 - (-4) = 5 + 4 = 9

(b) **Minimum possible value of $(p \times q)$:** To minimize the product $(p \times q)$, we want a negative product with the maximum possible absolute magnitude. This occurs by multiplying the largest positive value of one variable with the smallest (most negative) value of the other:

Min(p $\times q) = 5 \times (-4) = -20 \quad (\text{or } -5 \times 4 = -20)$

(c) **Subtracting minimum product from maximum difference:** We subtract -20 from 9:

- (-20) = 9 + 20 = 29

Final Answers: (a) 9, (b) -20, (c) 29

P10. Solution:

(a) **Divisibility Proof:**

- Since $a \mid b$, there exists an integer k_1 such that $b = k_1 a$.
- Since $b \mid c$, there exists an integer k_2 such that $c = k_2 b$.
- Substitute $b = k_1 a$ into the equation for c:

c = k$_2(k_1 a) = (k_2 k_1)a$

- Since k_1 and k_2 are integers, their product $k = x = k_2 k_1$ is also an integer.
- Thus, $c = ka$, which by definition means that a divides c $(a \mid c)$. The proof is complete.

(b) **Finding a set of values:** We need distinct non-zero integers a, b, c such that $a \mid b$, $b \mid c$, and $a \times b \times c = -64$. Let's test potential divisors of -64.

- Let $a = 2$.
- A divisor of -64 that is a multiple of 2 is $b = 4$ (since $2 \mid 4$, where $k_1 = 2$).
- Then, c must be: Thus, one possible set of values is $(a, b, c) = (2, 4, -8)$.

c = -64$\frac{}{a\times b=\frac{-64}{2\times 4}=\frac{-64}{8}=-8}$

- Now we check if the conditions are met:
  1. Are the integers distinct and non-zero? Yes, $\{2, 4, -8\}$ are distinct and non-zero.
  2. Does $a \mid b$? Yes, $2 \mid 4$ since $4 = 2 \times 2$.
  3. Does $b \mid c$? Yes, $4 \mid -8$ since $-8 = 4 \times (-2)$.
  4. Is the product -64? Yes, $2 \times 4 \times (-8) = -64$.